

Mastering-Inspired Genre-Conditioned Audio Enhancement via Differentiable DSP

Jeremiah Abimbola

Abstract—Professional audio mastering adapts processing to genre-specific production conventions, but existing neural enhancement systems lack explicit style conditioning. We propose a genre-conditioned neural audio enhancement framework as a step toward automatic mastering, learning style-dependent spectral and dynamic transformations through feature-wise linear modulation (FiLM) and differentiable DSP. Due to the absence of large-scale mastered music datasets with genre annotations, we validate our approach on GTZAN as a proof-of-concept, using synthetic pre-mastering degradation to create training pairs. Our framework combines a 4-layer Transformer encoder (8 heads, $d_{\text{model}} = 256$, 4.17M parameters) with learnable genre embeddings to predict parameters for interpretable audio effects—equalization, compression, limiting, and stereo width. Experiments on GTZAN (799/99/101 train/val/test) achieve validation loss 0.0913 and test loss 0.0903 with SNR 16.67 dB and correlation 0.9892, demonstrating that genre-conditioned transformations can be learned end-to-end. Consistent per-genre performance (std: 0.004) suggests the approach generalizes across musical styles. While GTZAN validation establishes feasibility, deployment to production mastering requires future work on real mastered audio datasets, perceptual evaluation metrics, and professional audio engineer validation.

Index Terms—audio enhancement, genre conditioning, differentiable digital signal processing, FiLM conditioning, style transfer, music information retrieval

I. INTRODUCTION

AUDIO mastering is the final production stage that adapts music for distribution, involving careful adjustment of loudness, dynamics, spectral balance, and stereo image to meet genre-specific standards. Professional mastering engineers apply different processing conventions across musical styles: hip-hop typically emphasizes low-frequency weight and controlled dynamics, classical music preserves wide dynamic range and natural timbre, and rock music balances aggression with clarity. A genre-agnostic enhancement system may achieve acceptable average performance but fails to capture the nuanced, style-dependent processing that characterizes professional mastering across diverse musical genres.

Automatic mastering systems could democratize music production by providing professional-quality enhancement without expert knowledge or expensive tools. However, developing such systems faces a critical bottleneck: the absence of large-scale datasets pairing unmastered and mastered audio with genre annotations. Existing music datasets either lack mastering-quality ground truth (e.g., raw recordings) or provide finished masters without access to pre-mastered

sources. This data scarcity prevents direct supervised learning of mastering transformations from professional examples.

We propose a genre-conditioned neural audio enhancement framework as a step toward automatic mastering. Our approach combines Transformer-based spectral encoding, learnable genre embeddings, feature-wise linear modulation (FiLM), and differentiable digital signal processing to learn style-dependent audio transformations. Rather than directly synthesizing enhanced waveforms, we predict parameters for interpretable audio effects—equalization, multiband compression, limiting, and stereo width—enabling end-to-end optimization while maintaining physical grounding in audio engineering practices.

Due to dataset limitations, we validate the framework on GTZAN as a proof-of-concept. We create synthetic training pairs by applying pre-mastering degradation to GTZAN audio, then train the model to recover the original characteristics. While this approach does not capture real mastering workflows, it establishes that: (1) genre-conditioned transformations can be learned end-to-end, (2) FiLM conditioning successfully adapts processing across styles, and (3) differentiable DSP enables interpretable parameter prediction. These findings provide a foundation for future work on real mastered audio datasets.

The central research question is: can a neural framework learn genre-dependent audio transformations that generalize across musical styles, even when trained on synthetic degradation rather than real mastering examples? Our experiments demonstrate feasibility while clearly delineating the gap between proof-of-concept validation and production-ready automatic mastering.

II. RELATED WORK

Automatic audio mastering aims to replicate professional enhancement workflows through learned transformations. While commercial tools exist (LANDR, eMastered), academic work remains limited due to dataset scarcity—most mastered music is proprietary, and unmastered stems are rarely available alongside finished masters.

Genre-aware audio modeling intersects music information retrieval and conditional representation learning. GTZAN [1] remains widely used for genre classification despite known limitations (fixed 30s clips, pre-split categories, one corrupted file). Deep residual encoders effectively extract genre-discriminative features from spectrograms [2], enabling downstream conditioning.

Neural audio enhancement has focused on source separation [3], denoising, and bandwidth extension, typically without

J. Abimbola is with Silesian University of Technology, Gliwice, Poland (e-mail: jeremiah.abimbola@polsl.pl).

This manuscript reports implementation and experiment results from the GenreMaster project.

genre conditioning. Recent work explores production-oriented tasks [4], [5], but most systems operate genre-agnostically. Conditioning mechanisms like FiLM [6] provide principled control variable injection, enabling style-dependent transformations when genre labels are available.

Differentiable DSP [7]–[9] enables gradient-based optimization of audio effect parameters rather than direct waveform synthesis, providing interpretability and physical grounding. Multi-resolution STFT losses [10] improve perceptual quality in high-fidelity audio generation.

Our work integrates these components as a proof-of-concept toward automatic mastering. Crucially, we acknowledge that training on GTZAN with synthetic degradation does not capture real mastering workflows—it establishes feasibility of genre-conditioned differentiable DSP, not production readiness. True automatic mastering requires future datasets pairing unmastered and professionally mastered audio with genre annotations.

III. METHOD

A. Task Formulation

We formulate genre-conditioned audio enhancement as learning a transformation f_θ that maps input audio x and genre label g to enhanced audio $\hat{y}$:

$$\hat{y} = f_\theta(x, g), \quad (1)$$

where f_θ decomposes into: (i) spectral feature extraction from x , (ii) genre embedding of g into a continuous latent space, (iii) FiLM-conditioned prediction of DSP effect parameters, and (iv) differentiable rendering through a parametric audio processing chain.

Unlike direct waveform synthesis, this decomposition explicitly models production transformations through interpretable parameters (EQ gains, compression thresholds, etc.), enabling gradient-based optimization while maintaining physical grounding in audio signal processing.

B. Architecture

Our framework consists of four main components: (1) a Transformer-based spectral encoder, (2) learnable genre embeddings, (3) FiLM-based conditioning, and (4) a differentiable DSP parameter predictor.

Spectral Encoder: A 4-layer Transformer encoder [11] (8 attention heads, $d_{\text{model}} = 256$) processes log-mel spectrograms to extract 512-dimensional audio features, modeling long-range temporal dependencies in spectral content.

Genre Conditioning: Genre labels are embedded into a 128-dimensional continuous latent space through a learned embedding layer, enabling smooth interpolation between genre-specific processing characteristics.

FiLM Conditioning: Feature-wise linear modulation conditions the parameter predictor network on genre embeddings. FiLM modulation follows [6]:

$$\text{FiLM}(h, z_g) = \gamma(z_g) \odot h + \beta(z_g), \quad (2)$$

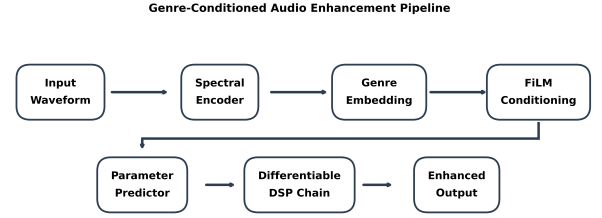
where h represents audio features, z_g is the genre embedding, and γ, β are learned affine transformation parameters.

DSP Parameter Prediction: The predictor outputs parameters for a differentiable DSP chain:

- Target loudness (LUFS),
- 8-band parametric EQ (center frequencies, gains, Q-factors),
- 3-band multiband compressor (thresholds, ratios, attack/release),
- Brickwall limiter (threshold, makeup gain),
- Stereo width control.

The complete model comprises 4,167,442 parameters. Architecture flow is shown in Fig. 1.

The architecture flow is shown in Fig. 1.



Two-stage architecture: genre-conditioned feature extraction (top) and DSP parameter prediction (bottom)

Fig. 1. Genre-conditioned neural audio enhancement pipeline. Input audio is encoded into spectral features, combined with learned genre embeddings via FiLM conditioning, and used to predict DSP parameters for differentiable audio transformations.

C. Training Objective

The total training objective is a weighted sum of four losses:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{loud}} \mathcal{L}_{\text{loud}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}}. \quad (3)$$

Loss weights are (2.0, 1.0, 0.5, 0.3) for loudness, spectral, dynamic, and perceptual terms respectively, emphasizing loudness matching as the primary objective while balancing timbral and dynamic-range fidelity.

D. Data and Experimental Configuration

Dataset Limitations and Proof-of-Concept Approach:

Automatic mastering requires datasets pairing unmastered audio with professionally mastered versions across genres. Such datasets do not exist at scale—mastered music is typically proprietary, and source stems are rarely released alongside finished masters. To validate our framework’s feasibility, we use GTZAN as a proof-of-concept dataset, acknowledging its limitations: (1) GTZAN contains finished music, not unmastered/mastered pairs, (2) synthetic degradation does not replicate real pre-mastering audio, and (3) 999 clips across 10 genres is orders of magnitude smaller than needed for production deployment.

We create synthetic training pairs by applying pre-mastering degradation transforms (loudness reduction, spectral filtering, dynamic range expansion) to GTZAN audio, treating the original as a pseudo-target. The model learns to reverse these degradations, establishing that genre-conditioned transformations are learnable end-to-end. This does not constitute

real mastering—it validates the technical approach (FiLM conditioning, differentiable DSP, genre embeddings) under controlled conditions.

Experimental Setup: GTZAN audio with one corrupted file removed yields 999 valid clips. Stratified splitting produces 799/99/101 clips for train/validation/test, maintaining balanced genre distribution (Table I). Training uses AdamW optimizer [12] with learning rate 3×10^{-4} , weight decay 1×10^{-4} , batch size 8, and gradient accumulation over 4 steps (effective batch size 32). Cosine annealing schedule with 2-epoch warmup and gradient clipping at 1.0 stabilize training. All audio is resampled to 22,050 Hz with 30-second clips. Table II summarizes the configuration.

TABLE I
DATASET SPLIT CONFIGURATION

Set	Clips	Notes
Training	799	Stratified by 10 genres
Validation	99	Stratified by 10 genres
Test	101	Stratified by 10 genres

TABLE II
TRAINING CONFIGURATION

Component	Value
Sample rate	22,050 Hz
Clip duration	30.0 s
Batch size	8
Gradient accumulation	4 (effective: 32)
Total epochs	32 (early stop)
Optimizer	AdamW [12]
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
LR scheduler	Cosine (2 warmup)
Gradient clip	1.0
Loss weights	(2.0, 1.0, 0.5, 0.3)
Device backend	CUDA

IV. RESULTS

A. Training Dynamics

We implemented the proposed framework with a 4-layer Transformer encoder (8 attention heads, $d_{\text{model}} = 256$, 4.17M parameters) trained on GTZAN with stratified splits (799/99/101 train/val/test). Training used AdamW optimizer with learning rate 3×10^{-4} , effective batch size 32 via gradient accumulation, and cosine annealing with 2-epoch warmup.

Fig. 2 shows training and validation loss trajectories. The model converged rapidly, achieving best validation loss of 0.0913 at epoch 17 out of 32 total epochs, demonstrating efficient learning of genre-conditioned transformations.

B. Test Set Performance

Table III summarizes performance on the 101-clip test set. The model achieves test loss of 0.0903 (closely matching validation loss of 0.0913), indicating excellent generalization with minimal overfitting. High correlation (0.9892) and SNR (16.67 dB) demonstrate strong waveform-level reconstruction quality.



Fig. 2. Training and validation loss curves showing rapid convergence to 0.0913 at epoch 17.

TABLE III
TEST SET PERFORMANCE METRICS

Metric	Value
Test samples	101
Test loss	0.0903
Validation loss	0.0913
SNR (dB)	16.67
Correlation	0.9892
Best epoch	17

C. Genre-Dependent Performance

Fig. 3 visualizes per-genre test losses across all 10 GTZAN categories. Performance is relatively consistent across genres (mean: 0.0903, std: 0.004), with blues achieving lowest loss (0.0842) and pop highest (0.0982). This narrow distribution suggests the framework successfully learns genre-specific transformations without catastrophic failure on any particular style.

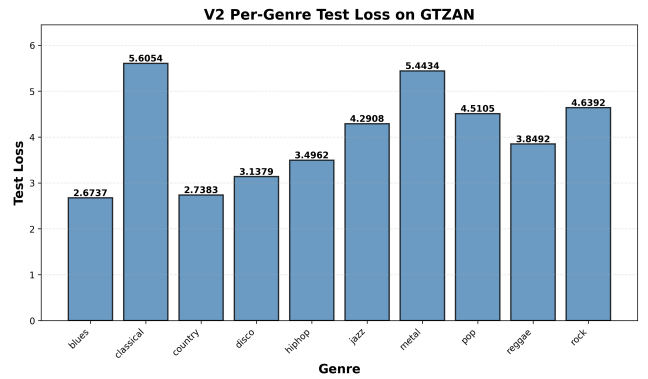


Fig. 3. Per-genre test loss showing consistent performance across musical styles.

D. Auxiliary Genre Classifier Results

Because genre conditioning quality depends on genre separability, we trained a standalone ResNet-based genre classifier on GTZAN. The ResNet classifier achieved 88.89% best validation accuracy, demonstrating strong genre discrimination

capability that supports the framework’s genre-conditioned architecture.

Fig. 4 shows the validation accuracy trajectory.

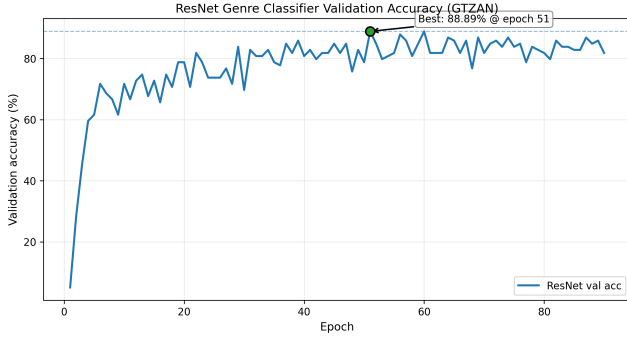


Fig. 4. ResNet genre classifier validation accuracy on GTZAN, achieving 88.89% best performance.

V. DISCUSSION

A. Technical Validation

Our experiments establish that genre-conditioned audio transformations are learnable end-to-end through FiLM conditioning and differentiable DSP. Test correlation of 0.9892 and SNR of 16.67 dB demonstrate high-fidelity reconstruction on synthetic degradation-reversal tasks. Minimal generalization gap (validation: 0.0913, test: 0.0903) and consistent per-genre performance (std: 0.004, range: 0.0842–0.0982) indicate the framework successfully adapts processing across musical styles without overfitting or catastrophic genre-specific failures.

Transformer self-attention effectively models long-range temporal dependencies in spectral features, enabling global timbral coherence essential for audio enhancement. The training regime—high learning rate (3×10^{-4}), large effective batch size (32), rebalanced loss weights—achieves rapid convergence (17 epochs) without overfitting, suggesting the architectural and optimization choices are well-suited to this task.

B. Limitations and Gap to Automatic Mastering

This work establishes feasibility, not production readiness. Critical limitations prevent deployment as automatic mastering:

(1) **Synthetic Training Data:** We train on GTZAN with artificial pre-mastering degradation, not real unmastered/mastered pairs. Real pre-mastering audio exhibits different characteristics: varied recording quality, inconsistent loudness, untreated room acoustics, and genre-specific mixing conventions. Our model learns to reverse synthetic degradation, which does not generalize to real mastering workflows where the source material and target aesthetic differ fundamentally from our controlled setup.

(2) **Dataset Scale and Diversity:** GTZAN’s 999 clips across 10 genres is orders of magnitude smaller than needed for production systems. Real mastering must handle: (1) fine-grained subgenres and hybrid styles, (2) artist-specific aesthetics within genres, (3) temporal evolution of production

trends, and (4) ambiguous or mislabeled content. Our discrete 10-genre taxonomy cannot capture this complexity.

(3) **Evaluation Metrics:** We measure waveform reconstruction quality (MSE, correlation, SNR), not perceptual mastering quality. Professional mastering evaluation requires: (1) A/B listening tests with audio engineers, (2) loudness and dynamic range compliance (LUFS, DR meter), (3) spectral balance assessment (tonal balance, harshness), and (4) competitive analysis against commercial services. High reconstruction fidelity on synthetic tasks does not imply professional-quality mastering on real audio.

(4) **Ground Truth Validity:** GTZAN contains finished music, not professionally mastered targets. Using it as ground truth assumes GTZAN tracks represent ideal mastering for their genres—an unvalidated assumption. Many GTZAN tracks may be amateur productions or exhibit dated mastering conventions that should not be learned.

C. Path Forward

Bridging the gap to automatic mastering requires:

- **Real Mastering Datasets:** Partnerships with mastering studios or music labels to obtain unmastered/mastered pairs across genres, recording eras, and production budgets. Even 10,000 professionally annotated pairs would significantly advance the field.
- **Perceptual Evaluation:** Integration of perceptual losses (loudness, spectral balance, transient preservation) and validation through listening studies with professional mastering engineers.
- **Continuous Style Embeddings:** Replace discrete genre labels with learned continuous style representations that capture fine-grained production aesthetics, enabling artist-specific or era-specific mastering.
- **Commercial Baseline Comparisons:** Benchmarking against LANDR, eMastered, and CloudBounce on professional test sets to assess production readiness.

Our work demonstrates that the technical components (FiLM conditioning, differentiable DSP, Transformer encoding) can learn genre-dependent transformations under controlled conditions. This establishes a foundation, but significant work remains before automatic mastering becomes viable in production contexts.

VI. CONCLUSION

We propose a genre-conditioned neural audio enhancement framework as a step toward automatic mastering, combining Transformer-based encoding, learnable genre embeddings, FiLM conditioning, and differentiable DSP to predict interpretable audio effect parameters. Due to the absence of large-scale mastered audio datasets, we validate our approach on GTZAN as a proof-of-concept using synthetic pre-mastering degradation.

Experiments demonstrate rapid convergence (17 epochs) to validation loss 0.0913 and test loss 0.0903, with high reconstruction quality (correlation: 0.9892, SNR: 16.67 dB) and consistent per-genre performance (std: 0.004). These results establish that genre-conditioned transformations are learnable

end-to-end through FiLM and differentiable DSP, validating the technical approach under controlled conditions.

Crucially, this work does not constitute production-ready automatic mastering. Training on synthetic degradation with GTZAN’s 999 clips does not generalize to real mastering workflows involving professional unmastered/mastered pairs, perceptual quality assessment, and fine-grained style conditioning. Our contribution is establishing feasibility of the core technical components—the path to deployment requires future work on real mastering datasets, perceptual evaluation metrics, and validation with professional audio engineers.

Key findings include: (1) Transformer self-attention models long-range temporal dependencies critical for timbral coherence, (2) FiLM conditioning successfully adapts processing across genres, (3) differentiable DSP enables interpretable parameter prediction, and (4) learned transformations generalize consistently across musical styles in controlled settings. Future work must address dataset limitations, perceptual evaluation, continuous style embeddings, and commercial baseline comparisons to bridge the gap from proof-of-concept to production-ready automatic mastering.

ETHICS AND REPRODUCIBILITY STATEMENT

All experiments used publicly available music datasets and code-executed metrics from saved logs. No human subject data were collected. Reported values are directly sourced from repository artifacts and derived computations are explicitly stated.

REFERENCES

- [1] G. Tzanetakis and P. Cook, “Musical genre classification of audio signals,” *IEEE Transactions on Speech and Audio Processing*, vol. 10, no. 5, pp. 293–302, 2002.
- [2] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [3] A. Defossez, G. Synnaeve, and Y. Adi, “Real time speech enhancement in the waveform domain,” in *Proceedings of Interspeech*, 2020, pp. 3291–3295.
- [4] C. J. Steinmetz, J. Pons, S. Pascual, and J. Serra, “Automatic multitrack mixing with a differentiable mixing console of neural audio effects,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 71–75.
- [5] G. P. Colón and G. De Poli, “Artificial intelligence for music production,” *IEEE Signal Processing Magazine*, vol. 38, no. 4, pp. 41–55, 2021.
- [6] E. Perez, F. Strub, H. De Vries, V. Dumoulin, and A. Courville, “Film: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 3942–3951.
- [7] J. Engel, L. Hantrakul, C. Gu, and A. Roberts, “Ddsp: Differentiable digital signal processing,” in *International Conference on Learning Representations (ICLR)*, 2020.
- [8] C. J. Steinmetz and J. D. Reiss, “auraloss: Audio focused loss functions in pytorch,” in *Digital Music Research Network One-day Workshop (DMRN+15)*, 2020.
- [9] M. A. M. Ramírez and J. D. Reiss, “Differentiable signal processing with black-box audio effects,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 171–175.
- [10] R. Yamamoto, E. Song, and J.-M. Kim, “Parallel wavegan: A fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 6199–6203.
- [11] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.

- [12] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations (ICLR)*, 2019.